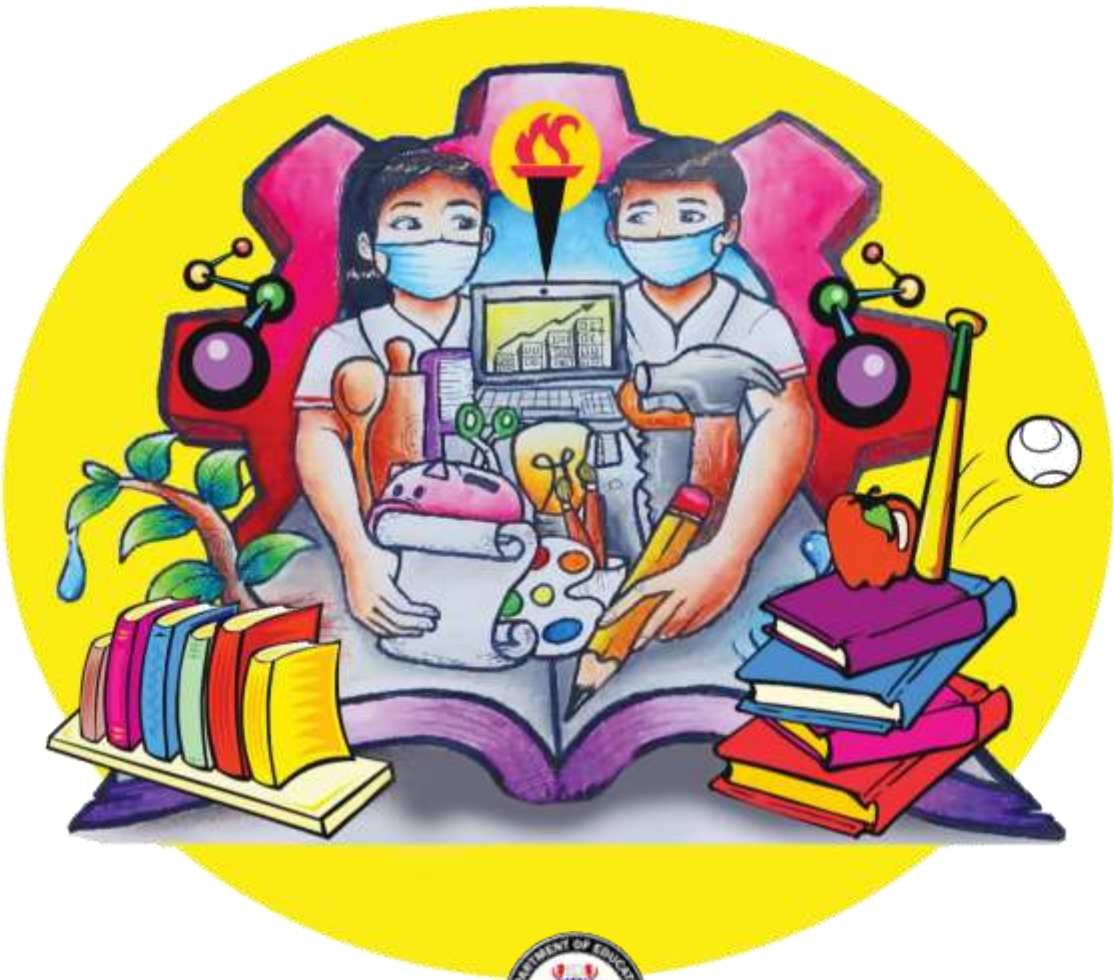
Senior High School

General Physics I

Quarter 2 – Module 2

Moment of Inertia



SLM

SELF -LEARNING MODULE

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Printed in the Philippines by

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Moment of Inertia

Introductory Message

For the facilitator:

Welcome to the General Physics I 12 Self-Learning Module on Moment of Inertia

This module was collaboratively designed, developed, and reviewed by educators both from public and private institutions to assist you, the teacher or facilitator in helping the learners meet the standards set by the K to 12 Curriculum while overcoming their personal, social, and economic constraints in schooling.

This learning resource hopes to engage the learners in guided and independent learning activities at their own pace and time. Furthermore, this also aims to help learners acquire the needed 21st-century skills while taking into consideration their needs and circumstances.

In addition to the material in the main text, you will also see this box in the body of the module:

Notes to the Teacher

This contains helpful tips or strategies that will help you in guiding the learners.

As a facilitator, you are expected to orient the learners on how to use this module. You also need to keep track of the learners' progress while allowing them to manage their learning. Furthermore, you are expected to encourage and assist the learners as they do the tasks included in the module.

For the learner:

Welcome to the General Physics I 12 Self-Learning Module on Moment of Inertia.

This module was designed to provide you with fun and meaningful opportunities for guided and independent learning at your own pace and time. You will be enabled to process the contents of the learning resource while being an active learner.

CONTENT STANDARDS

The learners demonstrate an understanding of:

1.Moment of Inertia
2. Angular position, angular velocity, angular acceleration
3. Torque
4. Static equilibrium
5. Rotational kinematics

PERFORMANCE STANDARD

Solve, using experimental and theoretical approaches, multi-concept, rich-content problems involving measurement, vectors, motion in 1D and 2D, Newton's Laws, Work, Energy, Center of Mass, momentum, impulse, and collisions

LEARNING COMPETENCIES

1.Calculate the moment of inertia about a given axis of single -object and multiple -object systems STEM_GP12RED - IIa -1
2.Calculate magnitude and direction of torque using the definition of torque as a cross-product STEM_GP12RED - IIa -3
3. Describe rotational quantities using vectors STEM_GP12RED - IIa -4

LEARNING OBJECTIVES:

1.Explain the moment of inertia and its effect on the motion of rotating bodies.
2.Explain the equivalence of mass in linear motion and moment of inertia in rotational motion.
3.Apply the concept of moment of inertia for single and multiple rotating object system.

INTRODUCTION

We are now in module 2 of the second quarter. Congratulations you made it! You did well in the previous module. What you will be learning today is just a continuation of what we just started in the previous lesson. This is just the second part of what we will be learning about rotational motion in this quarter.

The quantity to be presented in this lesson is the rotational counterpart of mass in Newton's Equation on his Second Law of Motion. You know very well about mass and its effect on the motion of a moving particle in linear motion. Truly, we cannot get away with mass when we talked of motion except in the kinematics of motion. Since we're done with kinematics let's move on to dynamics. Hence, this module was written to introduce you to one of its important aspects, the moment of inertia.

PRE-TEST

Name: ___________________________ Date: ____________
Year & Section: ____________________ Score: ___________

Direction: Encircle the letter of the correct answer**.**

1. Which of the following is the rotational analog of mass?
   A. Torque
   B. Angular velocity
   C. Moment of inertia
   D. Angular acceleration

2. Which of the following refers to the distance from the center of rotation where the mass of a body is believed to be concentrated without changing the moment of inertia of a system?
   A. Lever arm
   B. Center of mass
   C. Center of gravity
   D. Radius of gyration

3. Which has a greater moment of inertia?
   A. The short, fat man
   B. The tall, fat man
   C. The short, thin man
   D. A tall, thin man

4. The moment of inertia of a system depends on:
   A. mass of the body
   B. shape and size of the body
   C. distribution of mass from the axis of rotation
   D. all of the above

5. Kevin has a solid ball, solid disk, and a hollow hoop, with the same mass and radius placed at the same height on an inclined plane. He wants to roll each down from rest at the same time. Which do you think of the three that will hit the edge of the plane first?
   A. solid ball
   B. Solid disk
   C. Hollow hoop
   D. All of them will hit the edge at the same time because they have the same mass and radius.

6. What is the moment of inertia of a thin rod with a mass of 2 kg and a radius of 5 m from its axis through an end perpendicular to its length?
   A. 16.4 kg-m^2
   B. 16.5 kg-m^2
   C. 16.6 kg-m^2
   D. 16.7 kg-m^2

7. Which of the following is not true about the moment of inertia?
   A. It is also known as rotational inertia.
   B. It is represented by a capital letter M.
   C. It is the product of the mass of the particle with the square of its distance from the axis of rotation.
   D. All of the above.

8. What is the moment of inertia of a solid ball with a mass of 20 kg and a radius of 0.4 m?
   A. 2.9 kg-m^2
   B. 3.0 kg-m^2
   C. 3.1 kg-m^2
   D. 3.2 kg-m^2

9. What will happen if you extend your arms and legs while sitting on a spinning stool that is frictionless on its axle?
   A. Moment of inertia and angular velocity increases.
   B. Moment of inertia and angular velocity decreases.
   C. Moment of inertia increases but angular velocity decreases.
   D. Moment of inertia decreases but angular velocity increases.

10. A merry-go-round has a mass of 150 kg and a radius of 5 m. Two children with a mass of 15 and 20 kg ride on its rim. What is the moment of inertia of the system?
    A. 4,565 kg-m^2
    B. 4,625 kg-m^2
    C. 4,635 kg-m^2
    D. 4,735 kg-m^2

REVIEW OF THE PREVIOUS MODULE

What have you learned from the previous module? Right, we talked about the three rotational quantities and their kinematics. These quantities are angular position, angular velocity, and angular acceleration and its kinematics which described the relationship of these quantities with time.

PRESENTATION OF THE MODULE

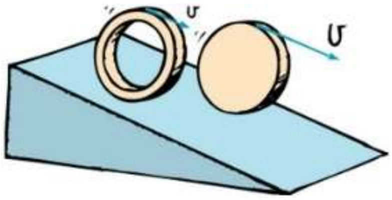
Look around you? Can you see an object capable of rotation? If yes, move closer then try to rotate that object. Does the object rotate? Was it hard or easy for you to rotate? So, what makes the object hard to rotate? Of course, it’s mass. Mass is a quantitative physical property that refers to an object's amount of matter. Therefore in the context of motion, the greater the mass the smaller the angular acceleration produced.

However, in rotational motion, we will use the term moment of inertia instead of mass. The term mass is no longer appropriate when we talk of rotational motion. Why? We will find out.

Activity 1:

Materials:

Look for any objects which resemble the following



- A solid sphere
- A hollow ring

1. Look for the above materials with the same mass and radius as much as possible.

2. Bring them together to an elevated ramp/ inclined plane at the same height and place them next to each other.

3. Roll the objects from rest at the same time. Observe.

Activity 2:

Material: pencil

Flip your pencil back and forth between your fingers. Compare the ease of rotation when you flip it about its midpoint vs flipping it about one of its ends. For a third comparison, rotate the pencil between your thumb and forefinger about the pencil's long axis. (so the lead is the axis). In which case is the rotation easiest? In this case, is the small rotational inertia consistent with the small radius? Explain.

ANALYSIS

(Activity 1)

1. Which object hit the bottom first? Which object is the last to reach the bottom of the plane?

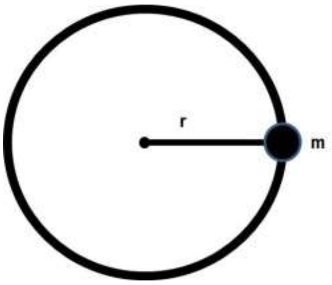
2. Explain why they don’t reach the end of the plane at the same time considering their similarity in mass and radius.

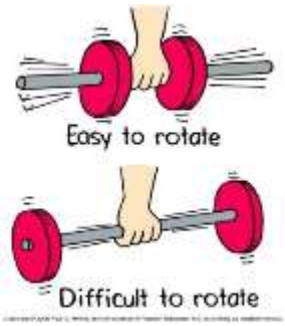
(Activity 2)

1. Which is easier to swing? Which is harder to swing? Why?
2. Considering, we are using the same ballpen. What could be the reason why the ballpen swings at different speeds holding it at different points across its length.

ABSTRACTION

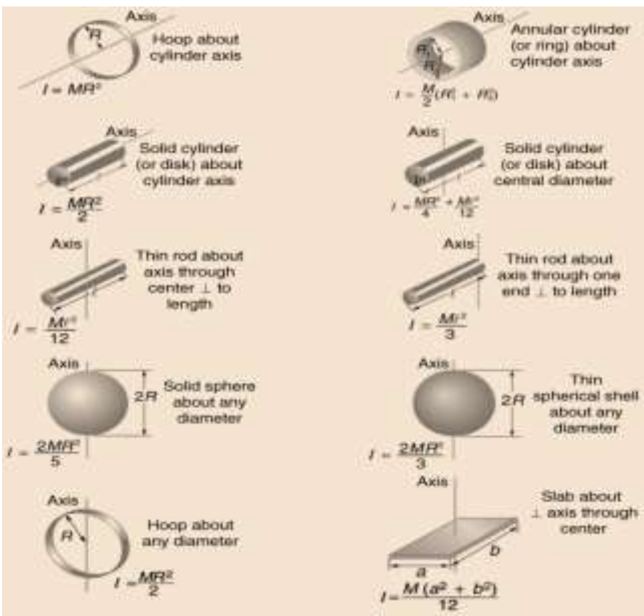
The moment of inertia also known as rotational inertia is a property of any rotating object which determines its resistance to angular acceleration. The moment of inertia is represented by the capital letter I which is the product of an object's mass and the square of its distance from the axis of a rotation expressed as $I = mr^2$.



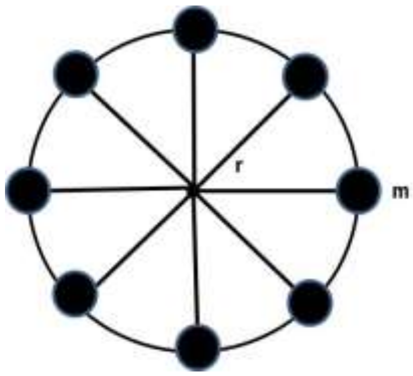


The property to resist changes in rotational state of motion is called **rotational inertia, or moment of inertia.** The small letter m represents the mass and r is the radius of a rotating object or simply the distance of an object from its center of rotation. The unit of moment of inertia is a kilogram. meter^2 (kg.m^2). From the equation above, it can be formally stated that an object with more mass and much bigger or a body farther from its axis of rotation has a greater moment of inertia and therefore the greater its resistance to angular acceleration. However, not only do mass and the length of the radius or an object's distance from its axis of rotation affect an object's moment of inertia.

Consider the figure below of some rotational inertia equation. Let's say that all of these shapes have the same mass and radius. Would they have the same moment of inertia? The answer is no, talking about the moment of inertia the shape of an object too shall be considered. The figure below shows the rotational inertia equations of the following shapes.



In a multiple-object system, the moment of inertia I am the sum of all the point masses which form part of the system that is $I = \sum mr^2$

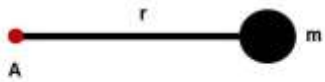


The radius of gyration (k) is the distance from the center of rotation where the mass of an object is believed to be concentrated expressed as;

$$k = \sqrt{\frac{I}{m}}$$

Sample Problem 1:

1. A ball with a mass of 2 kg is connected to a cord with a length of 0.5 m as shown in the figure below. Assumed the ball rotates from the axis of rotation A. What is its moment of inertia?



Solution:

$I = mr^2$

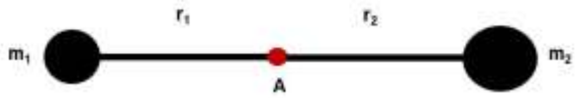
$= (2kg)(.5m)^2$

$= (2kg)(.25m^2)$

$= 0.5kg.m^2$

Sample Problem 2:

1. A 200 g ball, m1, and a 400-g ball, m2, are connected by a rod with a length of 120 cm. The mass of the rod is ignored. The axis of rotation is at the center of the rod. What is the moment of inertia of the balls about the axis of rotation?



Solution:

I = $\sum m_1 r_1^2 + m_2 r_2^2$

$= (0.2\text{kg})(0.6\text{m})^2 + (0.4\text{kg})(0.6\text{m})^2$

$= (0.2\text{kg})(0.36\text{m}^2) + (0.4\text{kg})(0.36\text{m}^2)$

$= (0.072\ \text{kgm}^2) + (0.144\text{kgm}^2)$

$= 0.216\text{kgm}^2$

Sample Problem 3:

1. What is the moment of inertia of a 5 kg long uniform rod with a length of 250 cm? The axis of rotation is located at one end of the rod.

Solution:

The formula of the moment of inertia when the axis of rotation is located at one end of the rod: See figure below.

$$I = \frac{\text{Ml}^2}{3}$$

$$\text{I} = \frac{(5\text{kg})(2.5\text{m})^2}{3}$$

$$\text{I} = \frac{(5\text{kg})(6.25\text{m}^2)}{3}$$

$$\text{I} = \frac{31.25\text{kg m}^2}{3}$$

$$\text{I} = 10.42\text{kg m}^2$$

APPLICATION

Rubric for scoring:

Criteria	Points
Complete solution with the correct answer.	10
The last two (2) major steps of the solution are incorrect.	8
Half of the solution is correct.	6
The first two (2) major steps of the solution are correct.	4
The first major step of the solution is correct.	2

1. A 3.5 kg rectangular thin plate has a length of 0.7 m and a width of 0.4 m. The axis of rotation is located at the center of the rectangular plate. What is the moment of inertia of the rectangular plate?

2. Find the moment of inertia of a disc of mass 3 kg and radius 50 cm about the axis passing through the center and perpendicular to the plane of the disc

POST-TEST

Direction: Write your answer in the space provided before the number.

1. The moment of inertia of a system depends on:
   A. mass of the body
   B. shape and size of the body
   C. distribution of mass from the axis of rotation
   D. all of the above

2. Which of the following is the rotational analog of mass?
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   C. 4,635 kg-m^2
   D. 4,735 kg-m^2

ANSWER KEY

Pretest	Post-test
1. C	1. D
2. D	2. C
3. B	3. D
4. D	4. B
5. A	5. B
6. D	6. D
7. B	7. A
8. D	8. D
9. C	9. C
10. B	10. B

Problems:

1. given m=3.5kg width = 0.4m length = 0.7m

Solution $I = m(L^2 + W^2)/ 12$

$= 3.5kg \times (0.7m)^2 + (0.4m)^2/ 12$

$= 3.5kg \times 0.65m^2/12$

$I = 0.19\ kg.m^2$

2. $I = ½ MR^2$

$= 1/2x3x(o.5)^2$

$= 0.375kg.m^2$

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Angela A. Silverio (2017). *Exploring Life Through Science: General Physics 1*:Phoenix Publishing House, Inc.

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